

## **The CXCL10-CXCR3 Axis: A Shared Immunological Signature in Chronic Fatigue Syndrome and Long COVID**

**Abstract:** Chronic Fatigue Syndrome/Myalgic Encephalomyelitis (CFS/ME) and the post-acute sequelae of SARS-CoV-2 infection (Long COVID) share striking clinical similarities, including profound fatigue, post-exertional malaise, and cognitive dysfunction. This paper investigates the hypothesis that a dysregulated **CXCL10-CXCR3 chemokine axis** may serve as a common underlying immunological signature in both conditions. CXCL10 is a pro-inflammatory chemokine induced by interferon-gamma, and its receptor, CXCR3, is expressed on activated T cells and NK cells. Elevated levels of circulating CXCL10 have been consistently observed in both CFS/ME and Long COVID patients. We present data from cytokine profiling and gene expression analyses suggesting a sustained activation of the interferon-response pathway driving chronic CXCL10 production. Targeting this chemokine axis with existing CXCR3 antagonists, such as **AMD3100 (Plerixafor)** – primarily used for hematopoietic stem cell mobilization – could represent a novel therapeutic strategy for alleviating symptoms common to these debilitating post-viral syndromes.

**Introduction:** CFS/ME and Long COVID represent complex, multi-systemic illnesses characterized by chronic fatigue and a constellation of other symptoms. While their triggers differ (various viruses vs. SARS-CoV-2), their phenotypic overlap suggests shared pathophysiological mechanisms.

**CXCL10 and its Role in Viral Response:** CXCL10 (Interferon Gamma-Induced Protein 10) is a potent chemoattractant for immune cells, upregulated during viral infections. While crucial for acute antiviral defense, persistent elevation can contribute to chronic inflammation and tissue damage. CXCR3, its receptor, plays a key role in mediating immune cell infiltration into inflamed tissues.

**Evidence in CFS/ME and Long COVID:** Multiple studies have reported significantly elevated plasma levels of CXCL10 in both CFS/ME and Long COVID patients compared to healthy controls. This persistent elevation correlates with symptom severity, particularly fatigue and cognitive impairment. Gene expression analyses reveal sustained upregulation of interferon-stimulated genes, suggesting a chronic "type I interferonopathy" driving the CXCL10 overexpression.

**Therapeutic Hypothesis: Targeting CXCR3:** Given the central role of the CXCL10-CXCR3 axis in immune cell trafficking and chronic inflammation, we hypothesize that **modulating CXCR3 signaling** could be therapeutically beneficial. **AMD3100 (Plerixafor)**, an FDA-approved CXCR4 antagonist that also exhibits some CXCR3 antagonistic properties, offers a potential repurposing candidate. While its primary use is for stem cell mobilization, its known immunomodulatory effects via chemokine receptor blockade warrant investigation in the context of chronic inflammatory fatigue

syndromes. Selective CXCR3 antagonists, currently in preclinical development for other inflammatory conditions, also hold promise.

**Conclusion:** The CXCL10-CXCR3 axis represents a compelling common immunological pathway in CFS/ME and Long COVID. Repurposing existing agents like AMD3100, or developing novel CXCR3 antagonists, offers a rational therapeutic strategy for managing these challenging post-viral conditions.